MOSFET, CS Amplifier and MOS Inverter



Basic Electrical and Electronics Engineering

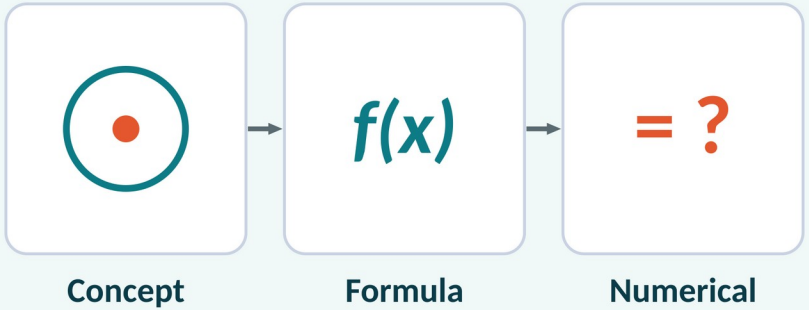
Unit 4: Transistors

Topic: n-channel enhancement MOSFET, characteristics, common-source amplifier, DC load line, MOS inverter

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Learning Outcomes

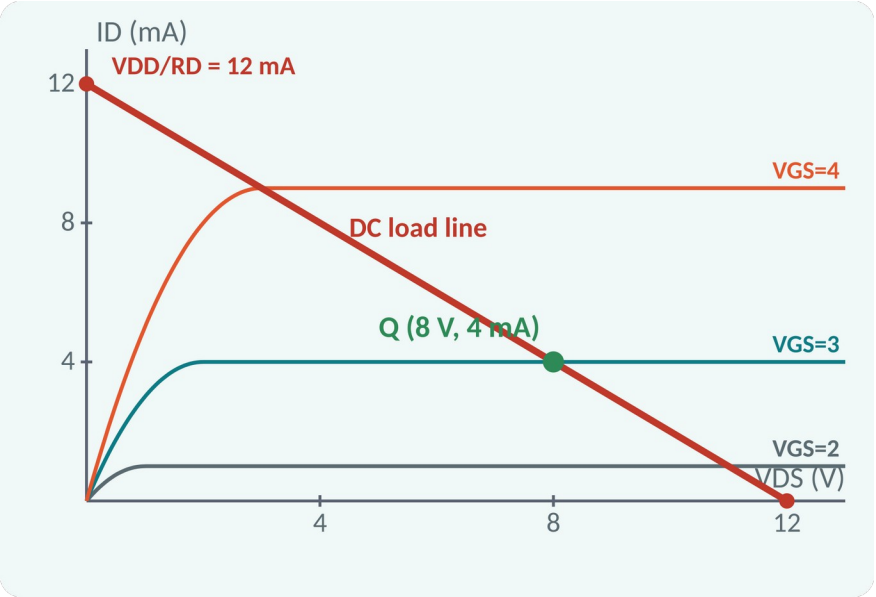
- Describe the structure and working of an n-channel enhancement MOSFET.
- Read the output and transfer characteristics.
- Find the Q-point using the DC load line.
- Explain the common-source amplifier and MOS inverter.



Linked to CO4
Describe the construction and operation of FET to operate as a switch.

Recap: What You Already Know

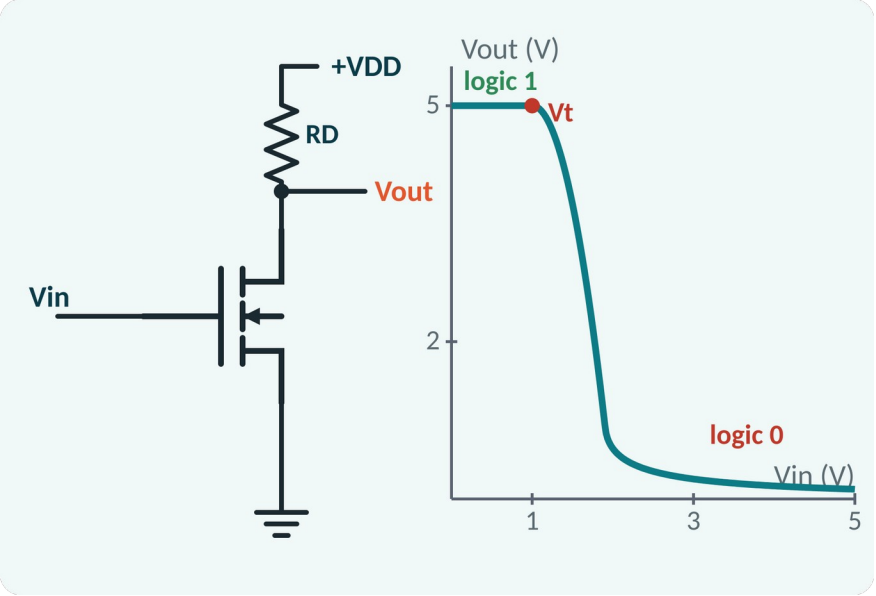
- BJT: current-controlled (IB controls IC).
- Load line: VCC = IC·RC + VCE.
- Q-point sets the DC bias.
- Switch: cut-off and saturation.



MOSFET: voltage-controlled (VGS controls ID); load line: VDD = ID·RD + VDS

Motivation: The Transistor of Modern Electronics

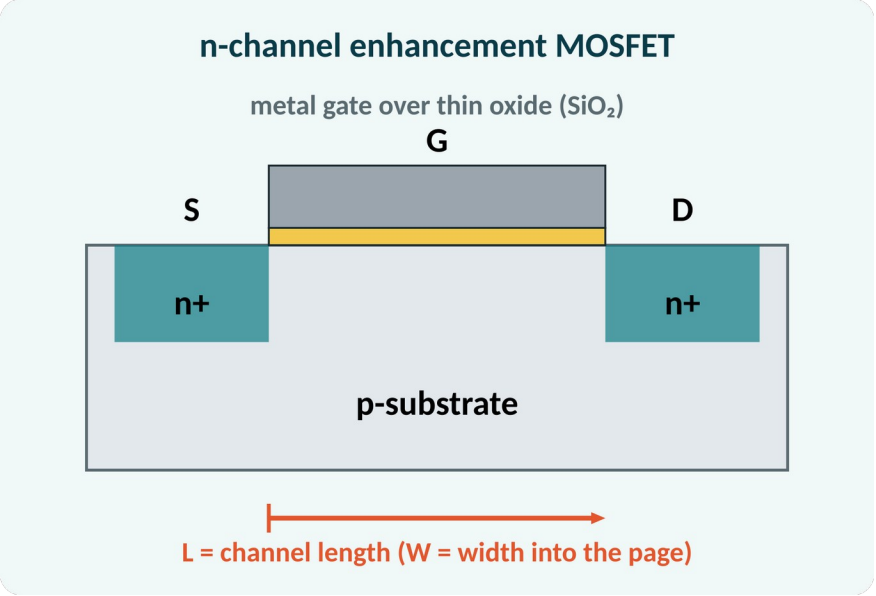
- Billions in a processor or phone.
- Very high input resistance; low power.
- Small, easy to integrate.
- Basic building block of logic gates.



MOSFET dominates ICs: amplifiers, memories, processors, power switches

Structure of the n-Channel Enhancement MOSFET

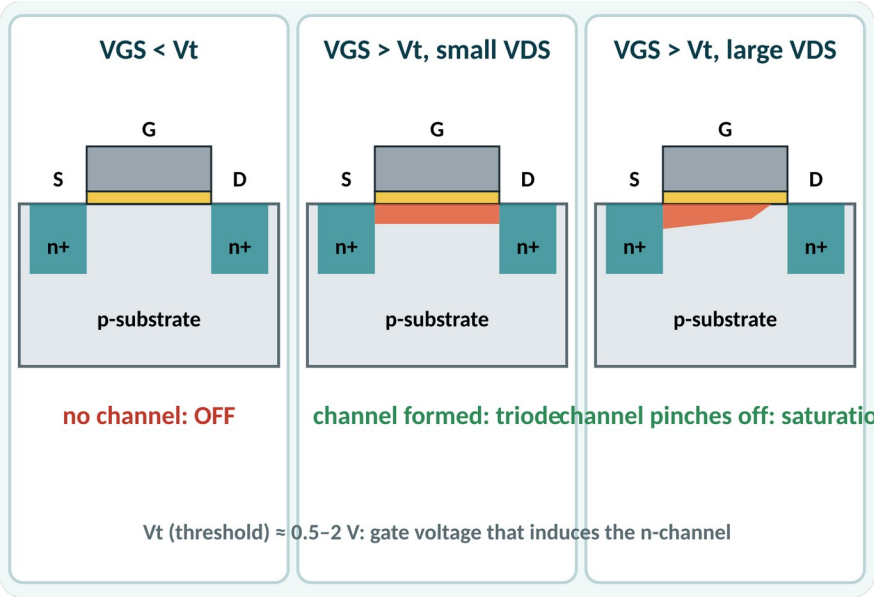
- p-type substrate (body).
- Two n+ regions: source and drain.
- Thin SiO_2 insulating layer over the channel region.
- Metal gate on the oxide.
- No channel exists at VGS = 0.



Terminals: Gate (G), Drain (D), Source (S), Body (B, tied to S); L channel length, W width

Working of the MOSFET

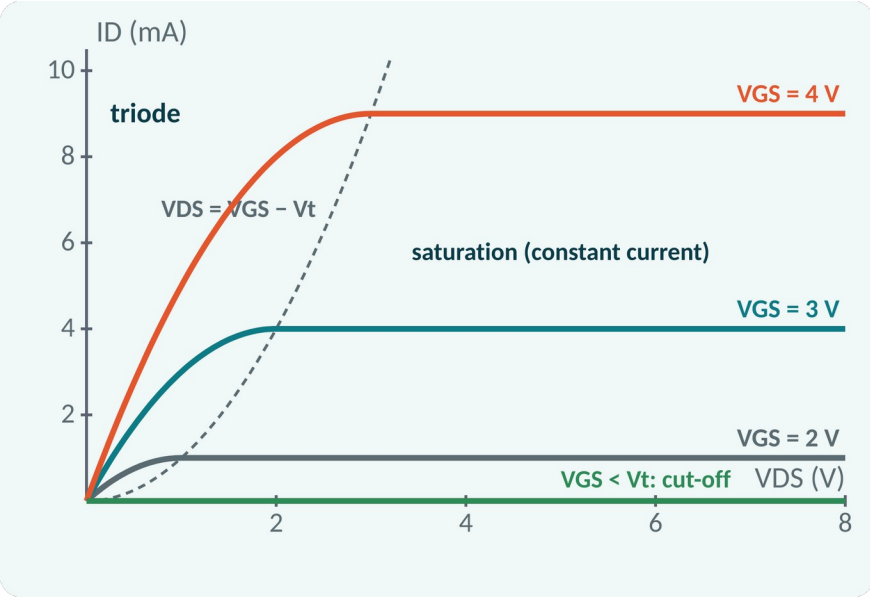
- VGS < Vt: no channel; ID ≈ 0 (OFF).
- VGS > Vt: positive gate attracts electrons: n-channel forms.
- Small VDS: channel acts as a resistor (triode).
- Large VDS: channel pinches off near the drain.
- Beyond pinch-off ID is nearly constant (saturation).



Channel induced when VGS > Vt; pinch-off when VDS = VGS − Vt

Output Characteristics

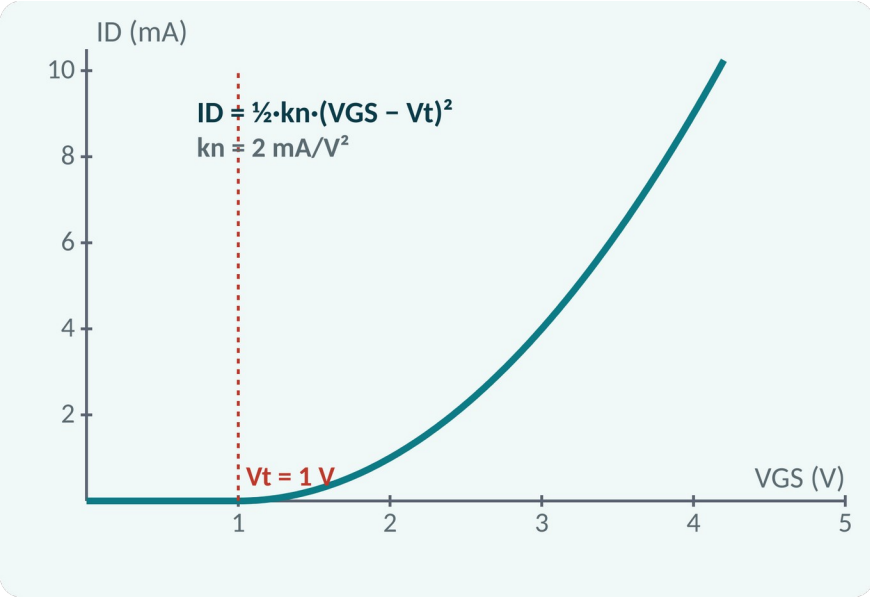
- ID against VDS for fixed VGS values.
- Triode region: ID rises with VDS (resistor-like).
- Saturation: ID nearly flat; used for amplification.
- Cut-off: VGS < Vt, ID = 0.
- Boundary: VDS = VGS − Vt.



Triode: VDS < VGS − Vt; saturation: VDS ≥ VGS − Vt; cut-off: VGS < Vt

Square-Law Equations and Transfer Curve

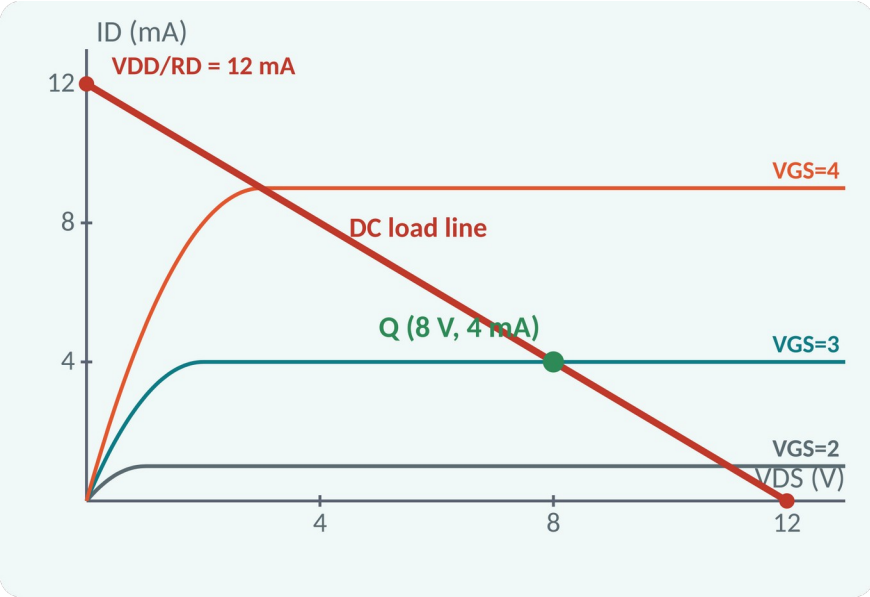
- Saturation: ID depends only on VGS.
- Square-law relation with threshold Vt.
- kn depends on W/L and process.
- Transfer curve: ID against VGS.



ID = ½·kn·(VGS − Vt)² (saturation); kn = μn·Cox·W/L (mA/V²)
Triode: ID = kn[(VGS − Vt)VDS − VDS²/2]

DC Load Line and Q-Point

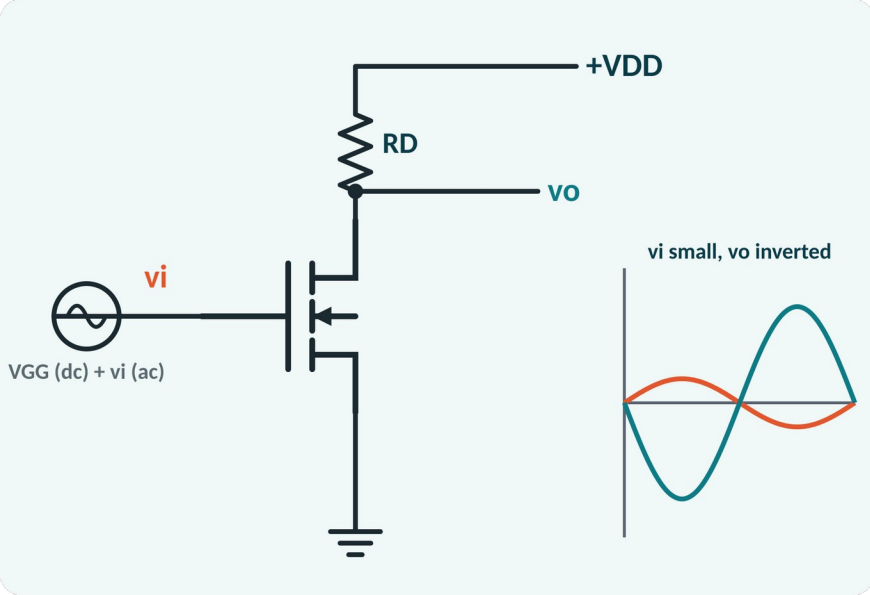
- Drain circuit: VDD = ID·RD + VDS.
- Load line joins (VDD, 0) and (0, VDD/RD).
- Q-point: intersection with the VGS curve.
- Amplifier: Q in the saturation region.
- Switch: operate at the two ends.



VDS = VDD − ID·RD; example VDD = 12 V, RD = 1 kΩ, VGS = 3 V: Q = (8 V, 4 mA)

Common-Source Amplifier

- Source is the common terminal.
- DC bias VGG sets the Q-point.
- Small input signal vi rides on VGG.
- Drain voltage varies inversely: vo opposite to vi.
- Gain larger than 1, phase 180°.



Bias: $I_D = \frac{1}{2} k_n (V_{GG} - V_t)^2$; $V_{DS} = V_{DD} - I_D R_D$

Small-Signal Gain

- Transconductance gm: change in ID per VGS.
- Output change: −gm·vgs·RD.
- Gain magnitude grows with RD and gm.
- Input resistance is very high.
- Output is 180° out of phase.

$$gm = \partial ID/\partial VGS = kn\cdot(VGS - Vt) = 2\cdot ID/(VGS - Vt)$$

$$Av = vo/vi = -gm\cdot RD$$

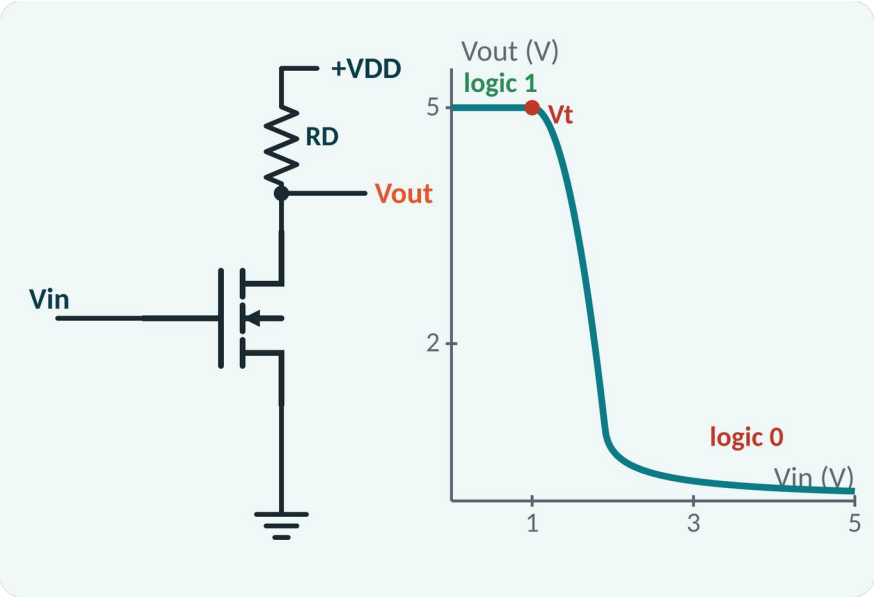
phase shift 180° (inverting)

input resistance very high (insulated gate)

$gm = kn(VGS - Vt) = 2\cdot ID/(VGS - Vt)$; $Av = -gm\cdot RD$
gm in mS (mA/V)

MOS Inverter

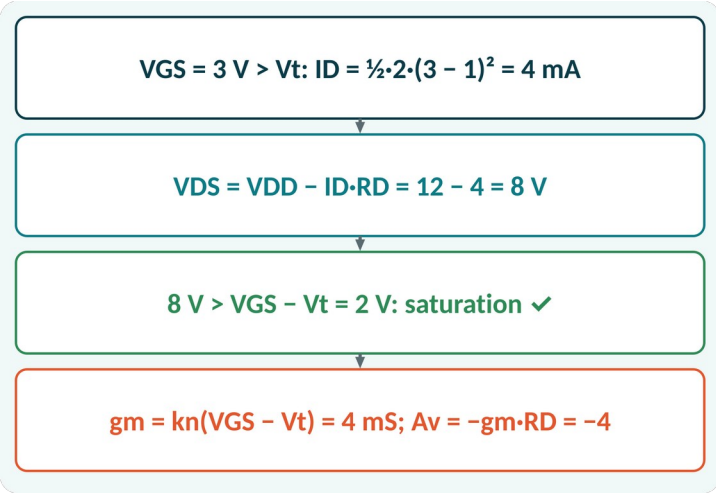
- NMOS with a drain resistor RD.
- Vin low (< Vt): MOSFET OFF → Vout = VDD (logic 1).
- Vin high: MOSFET ON, triode → Vout ≈ 0 (logic 0).
- Output is the logical NOT of the input.
- CMOS version replaces RD by a pMOS: no static power.



Vin < Vt: Vout = VDD; Vin = VDD: Vout = VDD − ID·RD ≈ VOL

Solved Example 1: Bias and Gain

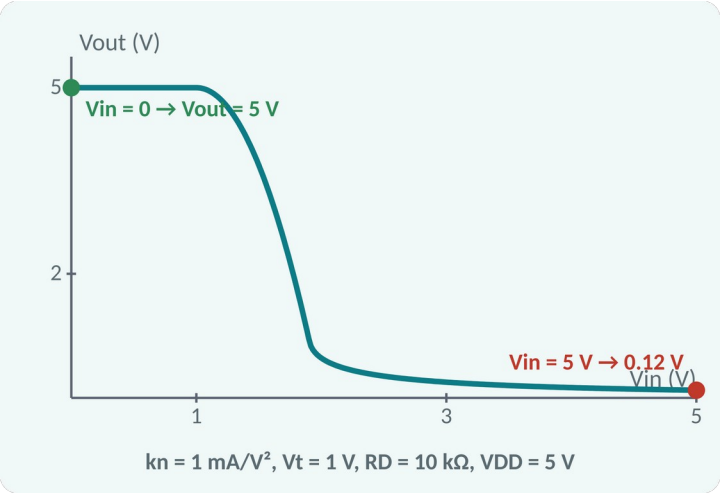
- NMOS: Vt = 1 V, kn = 2 mA/V², VDD = 12 V, RD = 1 kΩ, VGS = 3 V. Find ID, VDS, region, gm and Av.
- 1. ID = ½ × 2 × (3 − 1)² = 4 mA.
- 2. VDS = 12 − 4 × 1 = 8 V.
- 3. VDS (8 V) > VGS − Vt (2 V): saturation, as assumed.
- 4. gm = kn(VGS − Vt) = 2 × 2 = 4 mS.
- 5. Av = −gm·RD = −4.



Answer: ID = 4 mA, VDS = 8 V (saturation), gm = 4 mS, Av = −4

Solved Example 2: MOS Inverter

- VDD = 5 V, RD = 10 kΩ, Vt = 1 V, kn = 1 mA/V². Find Vout for Vin = 0 V and for Vin = 5 V.
- 1. Vin = 0 < Vt: MOSFET off, Vout = 5 V.
- 2. Vin = 5 V: assume triode: ID = 1·(4Vo − Vo²/2) mA.
- 3. Vo = 5 − 10·ID → 5Vo² − 41Vo + 5 = 0.
- 4. Vo = (41 − √1581)/10 = 0.124 V (< 4 V: triode ✓).



Answer: Vout = 5 V for Vin = 0; Vout ≈ 0.12 V for Vin = 5 V

Common Mistakes and Exam Tips

Common mistakes

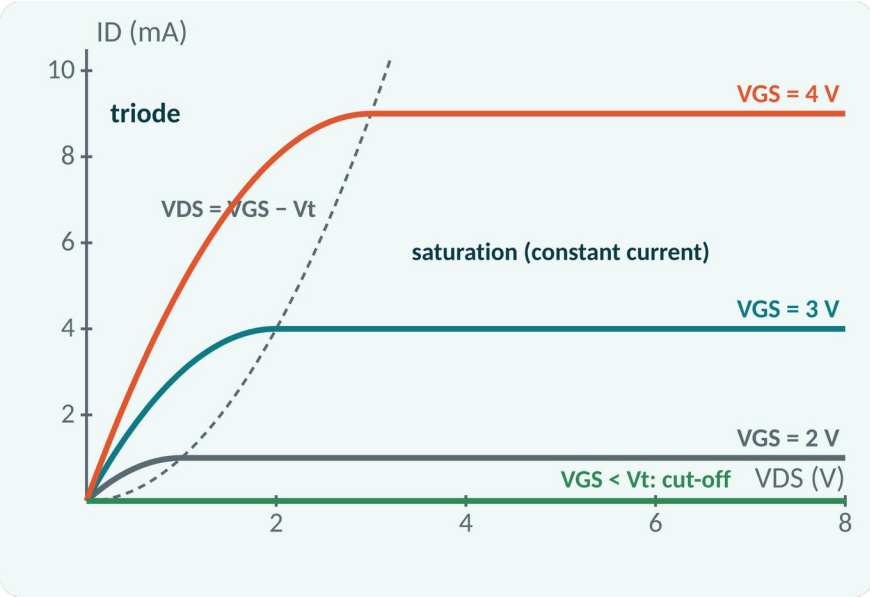
- Using the saturation formula in the triode region.
- Forgetting to check VDS ≥ VGS − Vt.
- Using VGS < Vt and expecting current.
- Treating the gate as drawing current.
- Forgetting the minus sign in Av.

Exam tips

- State the assumed region, then check it.
- ID = ½ kn (VGS − Vt)² only in saturation.
- Use ID = VDD-based load line for Q.
- Av = −gm·RD (inverting).

Summary

- Enhancement MOSFET: channel induced when VGS > Vt.
- Regions: cut-off, triode, saturation.
- Saturation: ID = ½·kn·(VGS − Vt)².
- CS amplifier: Av = −gm·RD (inverting).
- MOS inverter: Vin low → Vout high, Vin high → Vout low.



gm = kn(VGS − Vt); Av = −gm·RD; VDS = VDD − ID·RD

Quick Quiz

Q1. An n-channel enhancement MOSFET is OFF when:
(a) VGS > Vt (b) VGS < Vt (c) VDS > 0 (d) VDS = 0

Ans: (b)

Q2. Saturation ID for kn = 2 mA/V², VGS − Vt = 2 V:
(a) 2 mA (b) 4 mA (c) 8 mA (d) 1 mA

Ans: (b)

Q3. The common-source amplifier output is:
(a) in phase (b) 180° out of phase (c) 90° shifted (d) zero

Ans: (b)

Q4. gm for kn = 2 mA/V², overdrive 2 V:
(a) 1 mS (b) 2 mS (c) 4 mS (d) 8 mS

Ans: (c)

Q5. In an NMOS inverter, Vin = low gives:
(a) Vout low (b) Vout high (c) Vout = 0 always (d) undefined

Ans: (b)

References

1. **Boylestad Robert L. and Nashelsky Louis, Electronic Devices and Circuit Theory**
   11th Ed., PHI, 2016 (Units 3, 4, 5 textbook)

2. **H. Cotton, Electrical Technology**
   7th Ed., CBS Publishers, 2005 (reference)

Next lecture: Unit 5: operational amplifiers